

Are major cardiovascular events associated with anti-IL 12/23 agents?

Point: Major adverse cardiovascular events and anti-IL 12/23 agents

To the Editor: We read with interest the “tale of two meta-analyses” by Dommasch et al.¹ Briakinumab and ustekinumab are both human monoclonal antibodies targeting the same shared subunit (p40) of IL-12 and IL-23, belong to the same drug class, and demonstrate similar pharmacodynamics. Dommasch et al.¹ comment that briakinumab was withdrawn from further development because of concerns for cardiovascular safety, as well as for the possible excess risk of malignancy and serious infection. We emphasize that the rates for all side effects, including major adverse cardiovascular events (MACEs) in the controlled and uncontrolled periods are very similar, if not exactly the same, for both briakinumab and ustekinumab.²⁻⁵ Based on this evidence, an anti-IL-12/23 class effect cannot be excluded; consequently, a meta-analysis of drug class is valid.^{6,7}

Dommasch et al.¹ note the differences reported by both meta-analyses.^{4,5} The 95% CI for the Mantel-Haenszel risk difference (MHRD) reported by Ryan et al was -0.001 to 0.026 ($P = .12$).⁴ This is just 0.001 events per person-year less than if the results were statistically significant. By deferring from the P value and examining the level of certainty (1-p) in the results by Ryan et al, one comes to a confidence level of 88%, which is not considered to be statistically significant. In our meta-analysis,⁵ the 95% CI for the Peto Odds Ratio (PetoOR) are 1.07 to 16.75 ($P = .4$). This is only 0.07 away from being nonsignificant. The level of statistical certainty of our result was 96% and therefore our result, with the PetoOR, is classified as significant. Apart from arbitrary statistical P -value limits, the same effect is present in both meta-analyses. One could argue that the meta-analysis of Ryan et al demonstrated a trend that only needed a more appropriate statistical approach to be reported as significant.⁴ Notably, the MHRD method used by Ryan et al has been reported to be seriously limited in its ability to detect real differences, in that its confidence intervals are wide, giving poor statistical power.^{8,9} It has thus been reported to be unsuitable for meta-analysis of rare events, because it reduces the chance of real increases in rare adverse events being detected, with the subsequent possibility that patients continue to be exposed to harmful effects of some

medications.^{8,9} Overall, to overcome suggested limitations of both methods when used individually, it is important to conduct both to obtain the most thorough results, because they complement each other. For the clinician without extensive statistical experience, the bottom line is that, if a meta-analytical method “delivers” with regard to a rare adverse effect, it cannot be overlooked.

Dommasch et al report that the MHRD method is likely more appropriate for 2 reasons:

1. The exclusion of trials with zero events from the analysis: This reflects the previously described feeling that it is wrong to exclude any studies from a meta-analysis.^{8,9} However, if no events occur at all in a trial, the trial in itself conveys no information about the relative odds or risks of events between the two groups.^{8,9} A meta-analysis may be viewed as a weighted average of trial results, with weights reflecting the amount of information each study contains about the summary statistics. Allocating a trial with no information a zero weight is entirely appropriate and does not introduce bias.⁸
2. The PetoOR method used has been shown to inflate estimates of risk when groups are unbalanced: This has indeed been reported for imbalances between the numbers in the two groups of 8:1 or higher.^{10,11} All the studies included in these meta-analyses had an experimental to placebo ratio of 1:1 to 1:5; therefore, the suggestion by Dommasch et al is not supported by the evidence.

Dommasch et al comment that dropouts could have augmented the true risk for MACEs.¹ This is only a speculation that has to be tested in an individual patient data meta-analysis of randomized controlled trials, as we have previously suggested.⁵

The bottom line for clinicians is also that chronic plaque psoriasis is not a life-threatening disease. If it has a life-threatening component, it is due to the discussed possible increase in cardiovascular morbidity. The potential of anti-IL-12/23 biologic agents to further increase cardiovascular morbidity cannot be excluded given currently available evidence. Although tempting to theorize that these patients would have developed MACEs

anyway, caution is warranted due to the fact that the populations randomized to placebo exhibited no MACEs. To be on the safe side, based on limited available evidence, it could be more prudent to optimize those risk factors before initiating ustekinumab treatment.⁷ It could be more prudent to even consider ustekinumab as last treatment option for those chronic plaque psoriasis patients with known cardiovascular risk factors, especially given that anti-TNF treatment has been reported to have protective effect against MACEs.¹²

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Counterpoint: A tale of two meta-analyses revisited

To the Editor: We appreciate the interest of Tzellos and colleagues in our commentary regarding meta-analyses of major adverse cardiovascular events (MACEs) occurring in trials of briakinumab and ustekinumab, and we would like to emphasize several points.¹ Our aim was to demonstrate that meta-analyses of rare events using only study-level data are inherently limited, and thus the results of these studies should be interpreted with caution.